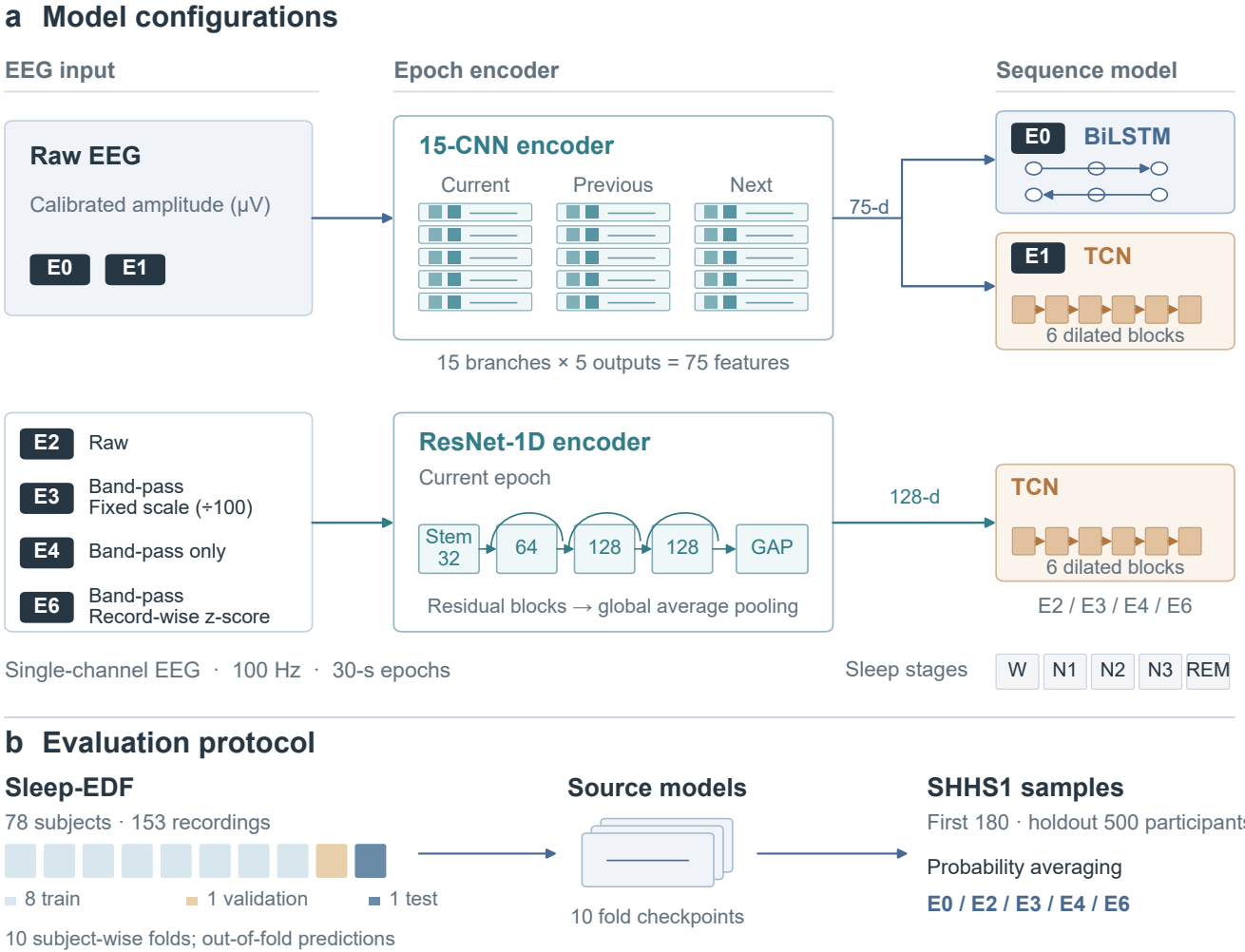

a Model configurations
EEG input
Epoch encoder
Sequence model
Raw EEG
Calibrated amplitude (μV)
E0
E1
15-CNN encoder
Current
Previous
Next
75-d
E0
BiLSTM
E1
TCN
6 dilated blocks
15 branches × 5 outputs = 75 features
E2
Raw
E3
Band-pass
Fixed scale (÷100)
E4
Band-pass only
E6
Band-pass
Record-wise z-score
ResNet-1D encoder
Current epoch
Stem
32
64
128
128
GAP
Residual blocks → global average pooling
128-d
TCN
6 dilated blocks
E2 / E3 / E4 / E6
Single-channel EEG · 100 Hz · 30-s epochs
Sleep stages
W
N1
N2
N3
REM
b Evaluation protocol
Sleep-EDF
78 subjects · 153 recordings
8 train
1 validation
1 test
10 subject-wise folds; out-of-fold predictions
Source models
10 fold checkpoints
SHHS1 samples
First 180 · holdout 500 participants
Probability averaging
E0 / E2 / E3 / E4 / E6